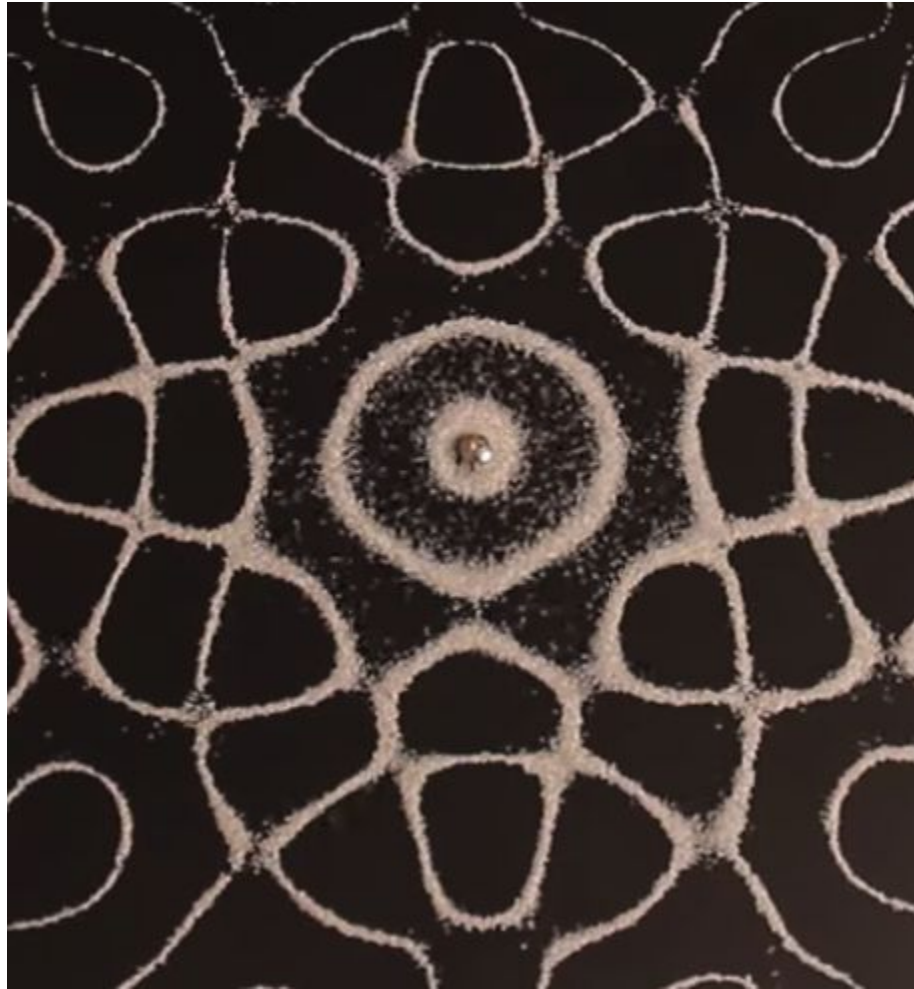




SYDE252 - W04A - System Response

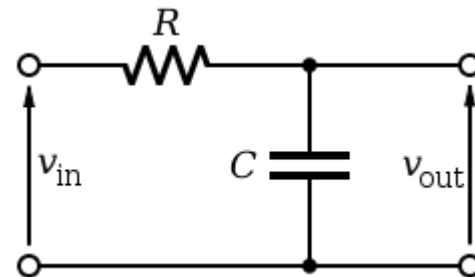
Dr. Matt Borland

For linear systems, the response of the system is the sum of ZIR and ZSR:

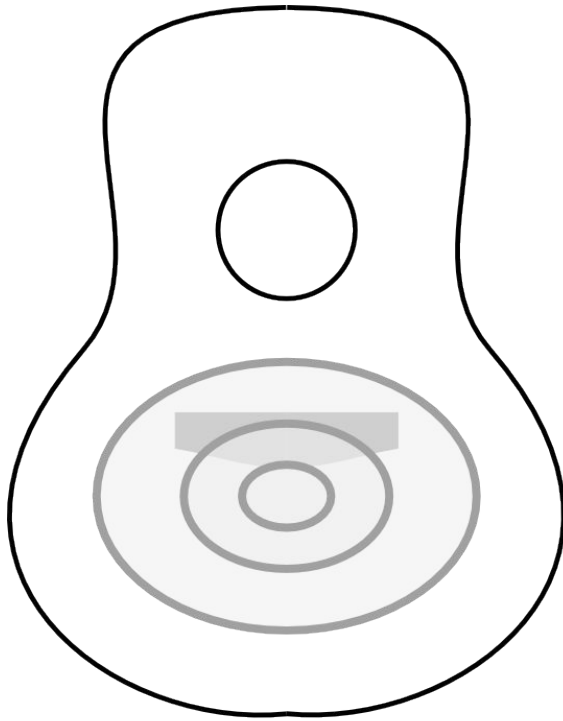
Total Response =
Zero-Input Response + Zero-State Response

The zero-input response is the system output when $x(t) = 0$.

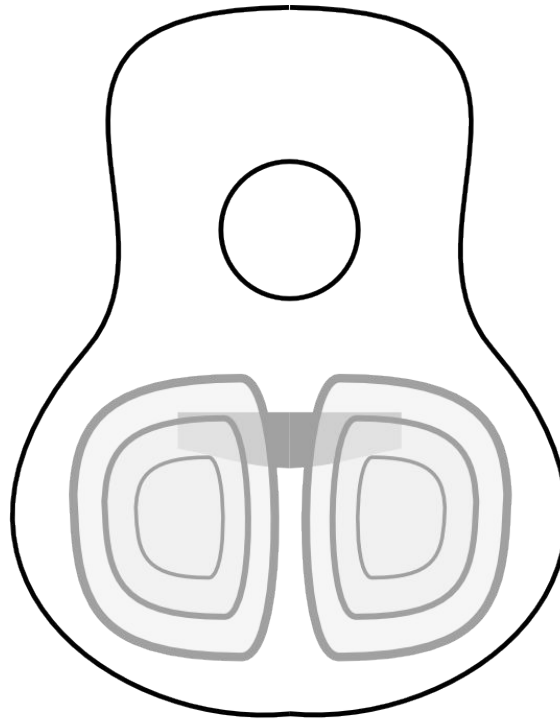
The zero-state response is the system output to the external input $x(t)$ when the system is in zero state, meaning the absence of all internal energy storages: that is, all initial conditions are zero.



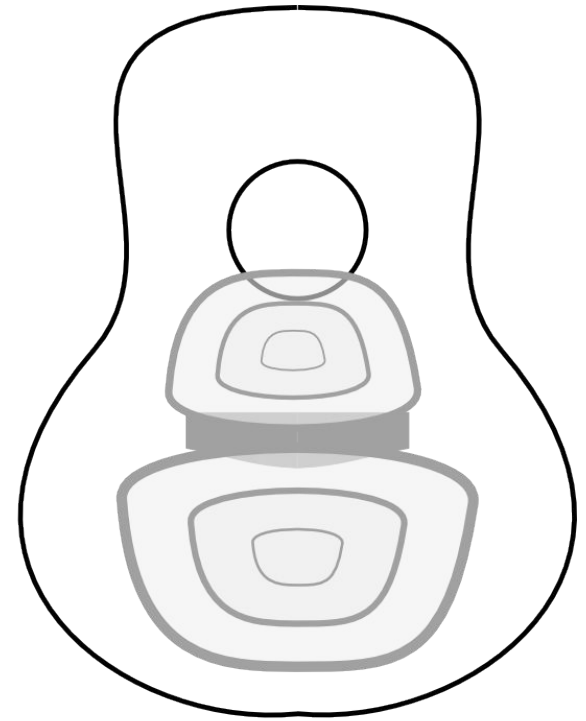
Modes in action!



Mode 1



Mode 2



Mode 3

<http://www.acs.psu.edu/drussell/guitars/hummingbird.html>

Modes in action!



The ZIR of a system is a linear combination of the modes of the system!

<https://youtu.be/wvJAgUBF4w>

<https://www.youtube.com/watch?v=-CZlrgg8syE>

Characteristic Modes

Lots of math, but this is pretty important stuff...

CHARACTERISTIC MODES

AAV Asine OPERATOR NOTATION

$$\frac{d^N y(t)}{dt^N} \rightarrow D^N$$

THIS IS JUST A SHORT FORM

LAST WE DAY
WE SAW THAT LINEAR
CONSTANT COEFFICIENT DES
ARE LINEAR

$$(D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N) \overset{\text{output}}{y(t)} \\ = (b_{N-M} D^M + b_{N-M+1} D^{M-1} + \dots + b_{N-1} D + b_N) x(t)$$

LET:

$$Q(D) = D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N$$
$$P(D) = b_{N-M} D^M + b_{N-M+1} D^{M-1} + \dots + b_{N-1} D + b_N$$

$$Q(D) y(t) = P(D) x(t) \quad 2.2$$

FOR OUR PURPOSES WE USUALLY ASSUME $M \leq N$

THE ZERO INPUT RESPONSE $y_0(t)$ IS THE SOLUTION
OF 2.2 WHEN $x(t) = 0$

$$Q(D) y_0(t) = 0$$

$$(D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N) y_0(t) = 0$$

A LINEAR COMBINATION OF $y_0(t)$ AND ITS DERIVATIVES
EQUALS ZERO AT ALL t !

CHARACTERISTIC MONES

2

THIS CAN ONLY BE TRUE IF $y_0(t)$ AND ITS DERIVATIVES ARE OF THE SAME FORM

→ NEED TO CANCEL OUT AT SOME POINT

$$\begin{array}{r} a(x^2(t)) \\ b(2x(t)) \\ c(2) \\ + d(0) \\ \hline 0? \\ \text{Doesn't work} \end{array}$$

$$(D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N) y_0(t) = 0$$

2.3

→ ONLY AN EXPONENTIAL FUNCTION CAN SATISFY THIS

$$e^{\lambda t} \leftarrow \text{THANKS EULER!}$$

SO LET'S ASSUME $y_0(t) = c e^{\lambda t}$ c & λ ARE CONSTANTS

$$D y_0(t) = \frac{d y_0(t)}{dt} = c \lambda e^{\lambda t}$$

$$D^2 y_0(t) = c \lambda^2 e^{\lambda t}$$

$$D^3 y_0(t) = c \lambda^3 e^{\lambda t}$$

SUB INTO 2.3 $c (\lambda^N + a_1 \lambda^{N-1} + \dots + a_{N-1} \lambda + a_N) e^{\lambda t} = 0$

THE NON TRIVIAL SOLUTION

$$\lambda^N + a_1 \lambda^{N-1} + \dots + a_{N-1} \lambda + a_N = 0$$

THIS IS OF THE SAME FORM AS $Q(D)$

EXCEPT D IS REPLACED WITH λ

∴ $Q(\lambda) = 0$ → IF WE FACTOR THIS ...

CHARACTERISTIC MODES

3

$$Q(\lambda) = 0 \quad \text{FACTORED} \dots$$

$$Q(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_N) = 0$$

λ HAS N SOLUTIONS: $\lambda_1, \lambda_2, \lambda_3 \dots \lambda_N$

ASSUMING λ_i ARE ALL UNIQUE

REMEMBER $y_0(t) = C e^{\lambda t}$ IS AN GENERAL SOLUTION OF 2.3

So 2.3 HAS N POSSIBLE SOLUTIONS

$$C_1 e^{\lambda_1 t}, C_2 e^{\lambda_2 t}, \dots, C_N e^{\lambda_N t}$$

IF WE ADD ALL OF THESE TOGETHER
WE GET BACK TO THE GENERAL SOLUTION

$$y_0(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} + \dots + C_N e^{\lambda_N t}$$

NOTICE $Q(\lambda)$ HAS NOTHING TO DO WITH
THE INPUT $x(t)$. $Q(\lambda)$ IS CALLED THE
CHARACTERISTIC POLYNOMIAL.

$Q(\lambda) = 0$ IS THE CHARACTERISTIC EQN
OF THE SYSTEM

$\lambda_1, \lambda_2, \dots, \lambda_N$ ARE THE ROOTS OF $Q(\lambda)$
CHARACTERISTIC ROOTS, EIGENVALUES, OR NATURAL FREQ'S

CHARACTERISTIC MODES

4

THE EXPONENTIALS $e^{\lambda_i t}$ ($i=1, 2, 3, \dots, N$)

ARE THE CHARACTERISTIC MODES
OR NATURAL MODES
OR MODES

OF THE SYSTEM.

THE ZERO INPUT RESPONSE OF A
SYSTEM, $y_0(t)$, IS A LINEAR
COMBINATION OF THE MODES OF
THE SYSTEM.

FOR LTIC SYSTEMS THE MODES MATTER!

MODES DETERMINE THE ZIR

MODES PLAY AN IMPORTANT ROLE IN
DETERMINING THE ZERO-STATE RESPONSE!

THE BEHAVIOUR OF A SYSTEM IS DICTATED
PRIMARILY BY ITS MODES

SEE BOOK FOR REPEATED ROOTS
COMPLEX ROOTS

EXAMPLE 2.1c FIND THE ZIR OF $(D^2 + 4D + 40)y(t) = (D+2)x(t)$
WITH $y_0(0) = 0$ AND $\dot{y}_0(0) = 16.78$

THE CHAR. POLYNOMIAL IS

$$Q(\lambda) = \lambda^2 + 4\lambda + 40 \quad \dots \text{SO FACTOR THAT?} \dots \text{HOW?}$$

LET'S USE B. 8-10
(AKA ... USE MAGIC)
(DON'T EXPECT MAGIC ON THE
MIDTERM)

$ax^2 + bx + c = 0$
QUADRATIC FORMULA?

YOU CAN GET THERE, BUT
NEED TO DO SOME COMPLEX
REFACTURING ...

$$\begin{aligned}\lambda^2 + 4\lambda + 40 &= (\lambda^2 + 4\lambda + 4) + 36 \\ &= (\lambda + 2)^2 + 6^2 \\ &= (\lambda + 2 - j6)(\lambda + 2 + j6)\end{aligned}$$

$$\lambda = -2 \pm j6$$

LET'S WRITE $y_0(t)$ THEN ... BOTH REAL + COMPLEX FORMS...

COMPLEX

$$y_0(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$

$$y_0(t) = C_1 e^{(-2+j6)t} + C_2 e^{(-2-j6)t}$$

WE COULD USE THE
I.C.'S + SOLVE
FOR C_1, C_2

OR REWRITE IN REAL FORM

To Go From THE COMPLEX TO REAL Form, ~~DO SOME MAGIC~~
USE Eqn. 2.8 From THE Book...

$$\underline{2.8} \quad y_0(t) = C_1 e^{(\alpha + j\beta)t} + C_2 e^{(\alpha - j\beta)t}$$

COMPLEX

CAN BE REWRITTEN AS A COSINE ... THANKS EULER!

$$y_0(t) = C e^{\alpha t} \cos(\beta t + \theta)$$

REAL

So APPLYING THIS TO

$$y_0(t) = C_1 e^{(-2 + 6j)t} + C_2 e^{(-2 - 6j)t}$$

GIVES US:

$$y_0(t) = C e^{-2t} \cos(6t + \theta)$$

WE HAVE I.C.'s $y_0(0) = 2$

$$\dot{y}_0(0) = 16.78$$

} APPLY & SOLVE FOR
 C & θ

EASY, RIGHT?!?

FIND $\dot{y}_0(t)$:

$$y_0(t) = C e^{-2t} \cos(6t + \theta)$$

$$\dot{y}_0(t) = -2C e^{-2t} \cos(6t + \theta) - 6C e^{-2t} \sin(6t + \theta)$$

KNOW BASIC
EXPONENTIAL &
TRIG
DIFFERENTIALS

APPLYING I.C.'s ... BOTH AT $t=0$! VERY CONVENIENT!

$$y_0(0) = 2 = C \cos(\theta)$$

$$\dot{y}_0(0) = 16.78 = -2C \cos(\theta) - 6C \sin(\theta)$$

RE-ARRANGING + SUB IN

$$1) \quad c \cos(\theta) = 2$$

$$6.78 = -2(2) - 6c \sin(\theta)$$

$$20.78 = -6c \sin(\theta)$$

$$2) \quad c \sin(\theta) = -3.463$$

From TRIG

$$\sin^2(\theta) + \cos^2(\theta) = 1 \quad \dots \text{SO ADD 1) + 2) + SQUARE, etc}$$

$$c^2 (\cancel{\sin^2(\theta)} + \cos^2(\theta)) = 2^2 + (-3.463)^2$$

$$c^2 = 16$$

$$c = 4$$

With $c=4$, WE CAN SOLVE FOR θ ...

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\text{SO} \quad \tan(\theta) = \frac{-3.463/4}{2/4} = -\frac{3.463}{2}$$

$$\theta = -\pi/3$$

$\therefore$ THE REAL FORM OF THE ZIR IS

$$y_0(t) = 4e^{-2t} \cos(6t - \pi/3)$$

LET'S TRY THIS IN OCTAVE... W4B_roots.m